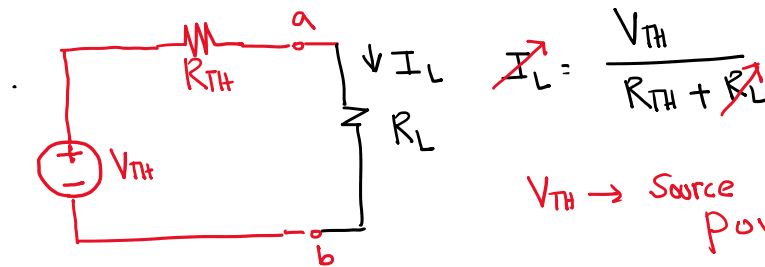
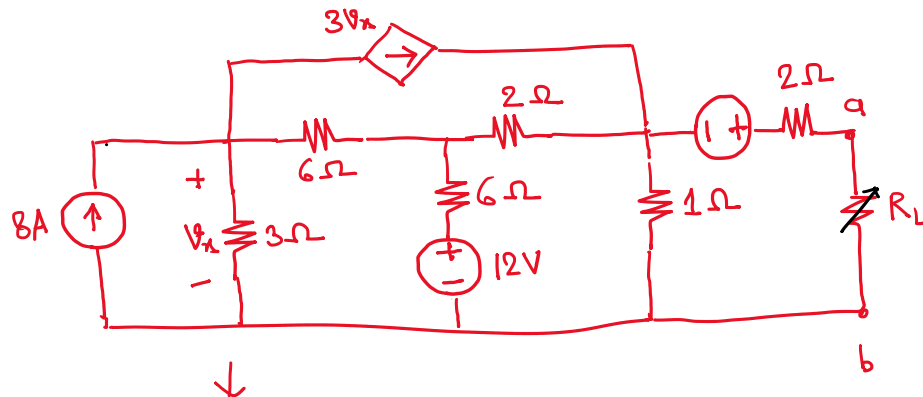


Maximum power transfer Theory



$V_{TH} \rightarrow$ Source that delivers power to the load

$$P_{del} = I_L V_{TH}$$

Power Absorbed by R_{TH} & R_L

$$P_1 = I_L^2 R_{TH} \quad , \quad P_2 = I_L^2 R_L$$

$$R_{Line} = 0$$



$R \propto L$. $(I^2 R_L \uparrow) \rightarrow$ Waste

Value of R_L at which P_2 will be maximum!

$$P_2 = I_L^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L \quad y = f(x)$$

$$\frac{dP_2}{dR_L} = 0$$

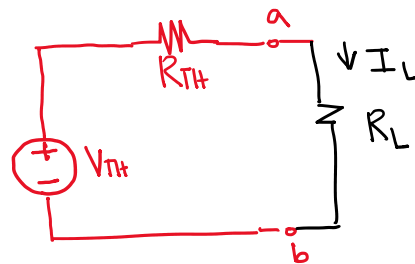
$$0 = R_{TH}$$

$$\frac{dy}{dx} = 0$$

$$\frac{dP_L}{dR_L} = 0$$

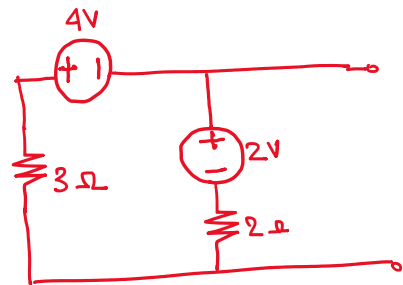
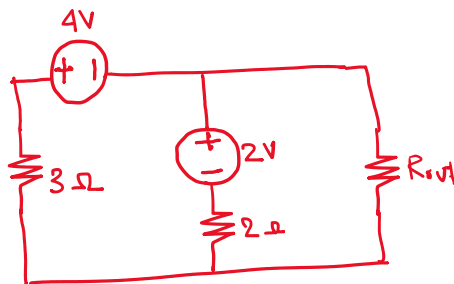
$$R_L = R_{TH}$$

$$\frac{dy}{dx} = 0$$

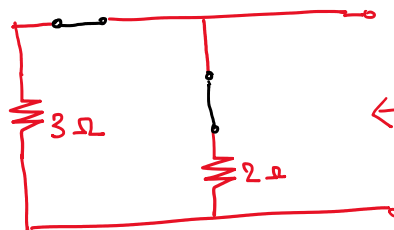


Condition for maximum Power Transfer
 $R_L = R_{TH}$

$$R_{out} = R_{TH}$$



V_{TH}, R_{TH}

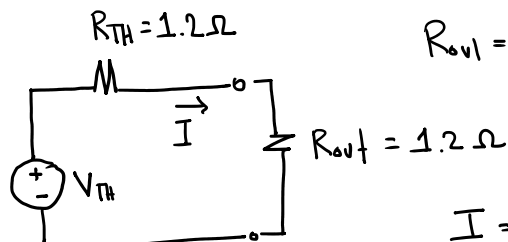


$$R_{TH} = 3 \parallel 2 \Omega = \frac{6}{5} \Omega$$

$$R_{out} = 1.2 \Omega$$

$$P_o = I^2 R_{out}$$

$$R_{out} = R_{TH}$$



$$I = \frac{V_{TH}}{2 R_{TH}} = \frac{V_{TH}}{2 \times 1.2}$$

$$= \frac{V_{TH}}{2.4}$$

$$P_o = \frac{V_{TH}^2}{4 R_{TH}^2} \times R_{TH}$$

$$P_o = \frac{V_{TH}^2}{4 R_{TH}}$$